A Case Study of the Epistemic Function of Citations - Implications for Citation-based Science Mapping

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Abstract

The use of citations as indicators of topical relatedness of publications is common in the algorithmic mapping of the structure of science. References to source documents, however, may serve a variety of epistemic functions, and hence represent rather different dimensions of topical relatedness. They may pertain to the research methods used, the empirical objects studied, the theoretical resources build on, the research questions pursued, or the external motivation for and relevance of the work. In this case study, we explore the diversity in topical dimensions along which publications are linked in citation networks, by coding the epistemic function of in-text citations. To construct an informative sample of publications, we first made a quality assessment of an existing field delineation of invasion biology and identified the core and the boundary of the field. We then sampled 9 publications from each layer to analyze the epistemic functions of citations. Our preliminary results for the core layer show the diversity of epistemic functions that underlie in-text citations and suggest substantial variation between studies. While current approaches to science mapping use citation links as generic indicators of relatedness, future algorithmic extractions that distinguish between epistemic functions of citations might greatly enrich science mapping.

Introduction

The use of citation links as indicators of topical relatedness is common in the algorithmic mapping of the structure of science (see e.g. Sjögårde & Ahlgren 2018). The resulting structures, consisting of interrelated groupings of publications are commonly interpreted as scientific fields (in global mappings), and topics or subfields (in field mappings). However, very few validation studies are done that inspect such algorithmically generated structures in order to explore their interpretation and relate them to a sociological understanding of the social and cognitive structures of science (Gläser et al. 2017). There is a tacit understanding that topics structures overlap and publications may reference each other along different topical dimensions (e.g. with regard to the methods used versus the empirical objects studied). However, current mapping approaches use citations as generic indicators of topical relatedness and apply algorithms that produce disjoint, non-overlapping clusters of publications, without systematically validating how these pragmatic choices affect the representation of fields and topics by these maps.

To improve our understanding of how such methodological choices may influence the topic structures represented by algorithmically generated science maps, we take a closer look at citations as the input data of the algorithmic mapping process, and investigate the diversity of dimensions of topical relatedness that citations may incorporate. We proceed in two steps that both involve an intellectual assessment, first of publications, then of citation links: In a first step, we investigate how the publications in a lexically delineated field data set relate to the cognitive structure of the targeted field, distinguishing four ordinal classes of degree of 'field focus.' This is to deepen our understanding of how the publications that we select for citation analysis in the next step relate to the targeted research specialty. We then construct a purposive sample of publications to represent different types of studies in the targeted research specialty and investigate the variation in topical dimensions of relatedness that citations entail by zooming in on their epistemic function. Early work in citation classification has primarily

examined the informative value of citations for assessing the impact of a cited publication (e.g. Moravcsik and P. Murugesan 1975, and Chubin & Moitra 1975). More recent work in citation classification has taken on further objectives, such as improving literature retrieval systems and automatic article summarization (Jha et al. 2017, Teufel et al. 2006). Our interest here is to consider the epistemic function of a citation in the citing text. Our coding scheme is inspired by recent work in the sociology of science that distinguishes different forms of epistemically mediated interdependencies in the collective knowledge production in the sciences (Gläser et al. 2018).

We present preliminary results, since the second half of the citation classification is ongoing. Our results about the degree of field focus for a random sample of 100 publications from the field data set are not only relevant for the construction of the purposive sample for this study, but also provide insights into the quality of the field delineation of invasion biology conducted by Held & Velden (*in preparation)*. We further report proportions of epistemic functions of in-text citations across different types of studies in invasion biology, and show how the proportion of cited sources that are located inside or outside the field differ by epistemic function.

Data and Methods

This is a case study, focused on publications in the field of invasion biology. Invasion biology is a research specialty that we are also studying through ethnographic observations and expert interviews to better understand research and sharing practices in this field. This background informs the sampling approach that we adopt in this study, and the decisions it entails, namely to distinguish publications by their degree of focus on the target field (invasion biology), and by study type, in order to stratify the sample by types of studies that are common in the field.

Field Data Set

To sample publications from the field of invasion biology we start from a bibliometric field data set that was generated using a lexical query to retrieve relevant articles and reviews from the Web of Science SCI and SSCI databases (Held & Velden, *in preparation*). The lexical query is an extension of the original query by Vaz et al. (2017) that was used in a previous citation-based mapping of the field of invasion biology (Held & Velden 2019). After constructing the direct citation network of the publications retrieved using the UTs of references, the final field data set is defined as the giant component of the network[i]. It consists of 53,524 publications published between 2000 and 2019.

Field Focus Coding

Since the manual coding of in-text citations is time consuming, we had to restrict ourselves to coding a small number of publications. To ensure the sample would represent studies in invasion biology well, we randomly sampled 100 publications from the field data set that were published in 2018, and classified them by the degree of their research focus on invasion biology ('field focus'). We defined a coding scheme that distinguishes four degrees of field focus, described in table 1. Coding decisions were taken in a step-wise process: first by determining whether a link to invasion biology existed or not[ii]. Then, if a link existed, by determining whether the link was strong or not. And finally, if the link was strong, by deciding whether the focus was singularly on invasion biology, or not. Following this decision tree, the 100 publications were coded independently by two coders. Basis for the assessment was the way that authors frame the contribution of their publication by composing title and abstract, choosing author keywords and selecting a journal to publish in. The inter-coder reliability, measured using a linearly weighted Cohen's Kappa, was 0.72[iii], a value which is commonly regarded as indicative of substantial agreement.

Table 1. Coding Scheme for Field Focus

Code	*Definition*
Core	Singular focus on invasion biology.
Boundary	Additional research focus, outside invasion biology.
Periphery	Explicit contribution to invasion biology is marginal.
Outside	No discernible connection to invasion biology.

Purposive Sample Construction for Citation Coding

We include in our sample core and boundary publications. Combined, we would argue, these types of publications define the field of invasion biology. We further take into account the type of study, distinguishing 'empirical' (observational, experimental), 'modeling'[iv], 'theoretical', and 'application-oriented' studies, in order to reflect the great variation in research approaches in invasion biology. We found the large majority of core and boundary publications to be empirical studies. Because theoretical studies were rare and occurred only among core publications, we omitted them from our sample for citation coding, and stratified the final sample of 9 core and 9 boundary publications by three study types: application-oriented, empirical, and modeling.

Citation Coding

Our classification scheme for capturing the epistemic function of in-text citations distinguishes between five epistemic functions that in-text citations may serve when referring to a source: clarifying the methodological approach (*method*), explicating properties of the empirical object (*empirical object*), drawing on or disputing theoretical insights (*theory*), defining and discussing the research problem (*research problem*), and referring to the external motivation or practical application of the work (*ext. interest*). If the context was ambiguous, we coded an in-text citation with up to two functions. We tried to use this option sparingly. The coding was conducted by two coders, one coding the core set of publications (see Seitz 2020), the other coding the boundary set, which is ongoing work. For resource reasons we could not afford double coding the entire sample. To ensure coding consistency, we performed selective double coding and review.

First Results

Field Membership

Our assessment of the field focus of randomly selected publications in the field data set finds that 71% (± 9%)[v] directly pertain to the field of invasion biology, either as core publications (49% ± 10%), or as boundary publications (22% ± 8%). Twenty-three percent (± 8%) are only peripherally related (e.g. simply mention the problem of invasive species as an ecological concern among other concerns). The remaining 6% (±5%) of publications are erroneously included, with no obvious relationship to the field.

Epistemic Functions of Citations

We observe considerable variation between the 9 core invasion biological studies in our sample. As figure 1 shows, the epistemic functions of in-text citations vary between studies, showing strong variation even among studies of the same study type. The most suggestive difference between study types seems to be that application-oriented studies tend to have more in-text citations that relate to external interest than empirical or modeling studies. Overall, in the

aggregate, external interest and theory related in-text citations are fewer than empirical object, method or research problem related citations.

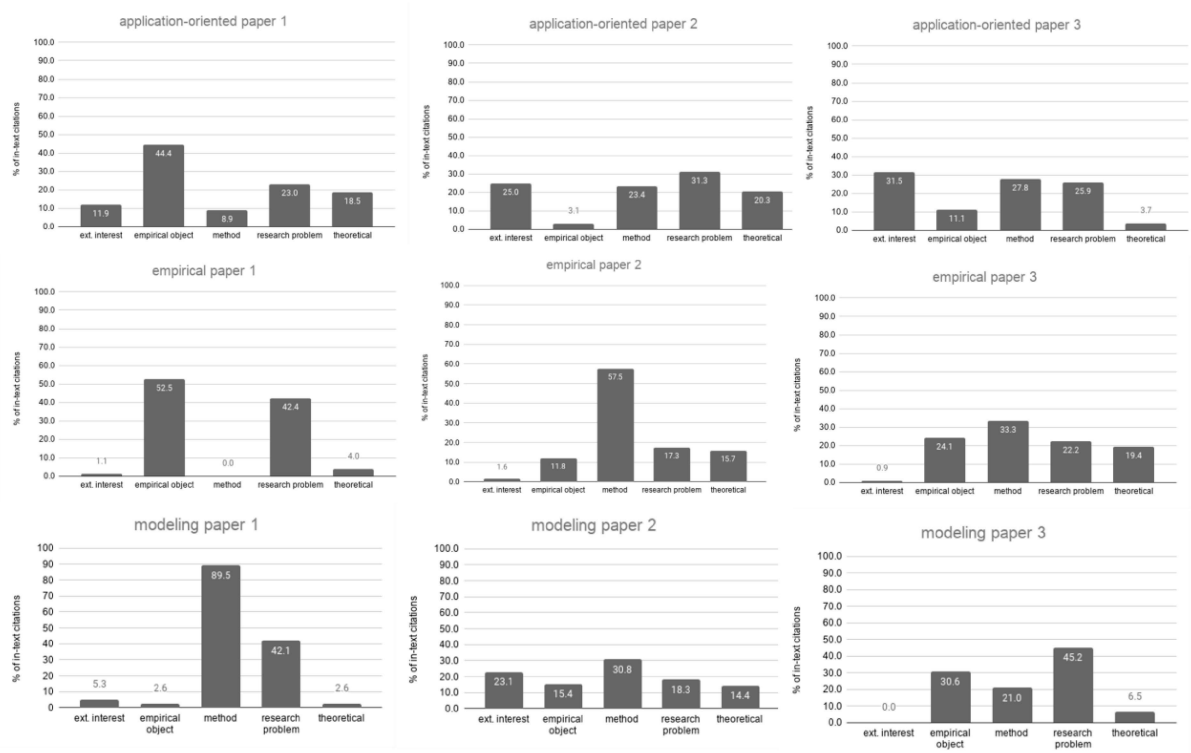


Figure 1. Epistemic function of in-text citations for each of the 9 publications in the core invasion biology sample.

Figure 2 shows the extent to which in-text citations in the 9 core invasion biology publications refer to publications inside versus outside the field data set. Only cited publications carrying a WoS UT and published within the time window of the field data set (2000-2019) are considered. Of the 568 references, 354 references in the 9 sample articles fulfill this criterion, corresponding to 571 in-text citations. Sources outside the field data set dominate among method related citations, whereas citations related to research problem, theory, or application aspects predominantly refer to publications within the field data set. For empirical object related citations the proportion of internal versus outside sources is almost even.

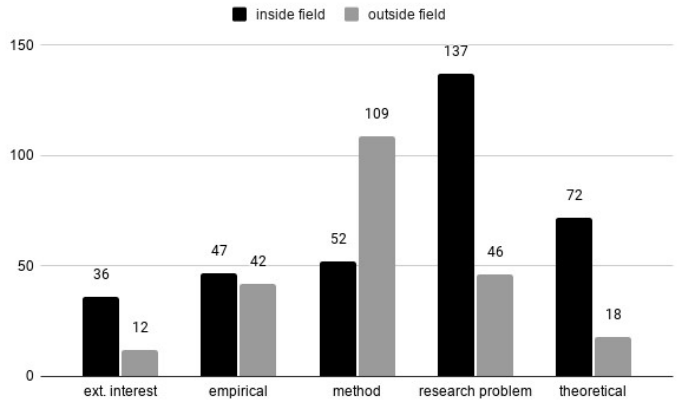


Figure 2. Proportion of in-text citations to publications inside versus outside the field data set by epistemic function (for subset of publications indexed by WoS and published in 2000-2019)

Discussion

How to get field delineation right is a critical issue in the bibliometric study of scientific fields (Zitt et al. 2019). Our coding of field focus is one possible approach to validate results and reveals that an estimated 71% of the lexically delineated field data set consists of publications that can be considered to represent research in invasion biology. Since this kind of validation exercise is rare, there is limited data to compare our result to. Haunschild et al. (2018) probed the purity of an algorithmically generated field classification by intellectual inspection of a random sample of 123 publications. They found roughly 50% of publications in the sample to belong to the target research specialty of 'common water splitting'. The decision criteria for whether a publication is considered to belong to the target field are not made explicit in their publication. Our approach differs in that we define a more fine-grained classification along with coding guidelines that help distinguish between those classes.

The process of developing the coding guidelines forced us to make explicit and agree on what, for us, defines field membership of a publication. By distinguishing between core and boundary publications, we found that a little more than 2/3rd of the field to consist of publications with a singular focus on invasion biology, and a little less than 1/3rd of the field to consist of publications that share a research focus on invasion biology with a research focus on a neighboring field. Importantly, some of the publications that were classified as singularly focused on invasion biology, simultaneously contribute to a field that can be interpreted as 'pervasively' overlapping. For example, invasion biological studies that examine the interaction of an invasive species with a potential biological control agent to be used in invasion control can be simultaneously regarded as studies in the field of pest control. This observation underlines the poly-hierarchical structure of scientific fields (Havemann et al. 2017).

The results of coding the epistemic functions of in-text citation in the 9 core invasion biology papers are intriguing. That application-oriented studies would refer more strongly to an external interest such as the motivation for the study and the application relevance of its result, seems plausible. The strong within group variation in the epistemic function of in-text citations is striking. It points to shortcomings of the high-level study classification that we define and points to a diversity of subtypes. Take e.g. the variation in the proportion of method related in-text citations among empirical studies. Empirical paper 1 constitutes a classical field study reporting a small-scale observational survey of species occurrences. It has no method related in-text citations. By contrast, empirical paper 2 uses genetic analysis to determine invasion success factors and proposes a methodological innovation, criticizing previous analyses for a methodological deficiency. In this paper more than 50% of in-text citations relate to methods.

If we look at the epistemic function of citations to the concurrent literature inside versus outside the field, this case study further provides a window into the epistemic interdependencies between fields. Our (small sample of) data suggests that invasion biology is highly reliant on method-related knowledge published in other fields, followed by knowledge about empirical objects. Knowledge pertaining to the research problem, external interest, and theoretical assumptions, on the other hand, is primarily referenced by citing publications from inside the field. Whether this pattern holds for other fields is an open question.

Conclusions

We provide empirical evidence for the diversity in topical dimensions of relatedness that are represented by in-text citations. We find great variation in the epistemic functions that drive citations to other publications within the field versus outside the field. The current practice in science mapping of using citations generically as signals of topical relatedness conflates these different dimensions that could be highly informative for science mapping, if only these different dimensions were algorithmically accessible for automated extraction.

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[i] Held &Velden (2019, in preparation) disregard publications outside the giant component in order to increase the precision of the field delineation.

[ii] At times, publications picked up by the lexical query do not relate to invasion biology. This happened when a publication uses a term in the abstract that is part of the lexical query in order to signal invasiveness (such as 'introduced'). To be relevant, the lexical query requires such terms to be combined with another relevant term (such as 'species'), however it does not restrict the order of terms and permits a two word distance between terms, which can lead to irrelevant matches.

[iii] After the inter-coder assessment, one item in the coding guidelines was tweaked one last time in order to increase the internal consistency of the guidelines. The change meant to categorize studies that have an invasive species as empirical object, but do not articulate a link to an invasion biological research question, as peripheral – in agreement with the weight given in our qualitative classification approach to the framing of studies. This resulted in the reassignment of three studies that focus on invasive species from boundary to peripheral. We did not re-assess inter-coder reliability after this change.

[iv] Studies that use computational or statistical modelling approaches to predict the distribution of invasive species or factors that influence mechanisms of invasive processes.

[v] Based on a 95% confidence interval for the estimate of a population proportion.